Supplement file 1

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Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

Figure S8. Proportion of domains covered by each reviewed tool

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In this supplement, we describe in greater detail the methods and results of our review process and Delphi survey and report additional findings related to the evaluation of each assessment tool.

As stated in the main manuscript, the framework to evaluate the assessment tools was developed based on the domains and subdomains indicated by the Comparative Effectiveness Research (CER) Special Interest Group (SIG) as the main critical elements that can threat to the validity of non-randomized studies (NRS) of medications.

Delphi Survey

To collect the domains and to ultimately develop the evaluation framework, we performed a survey following the Delphi methodology. This is an iterative multistage process that uses rounds of questionnaires to reach a consensus among content experts [Hasson et al. 2000]. The survey targeted all members of the SIG of the International Society for Pharmacoepidemiology CER. These experts were informed about the survey during an online meeting hold on July 1, 2019. During the meeting, study protocol and Delphi survey were presented to the group, addressing that answering to the survey would have constituted consent to participate in the study. After the online meeting, the study protocol was e-mailed to the members of CER SIG.

On August 23, 2019, a link to a self-administered online questionnaire was sent by email to 48 CER SIG members. In the first page of the survey, a cover letter outlined the purpose of the study and assured the participants of anonymity in processing the results. The online questionnaire could have only been submitted once per person. The experts had three weeks to answer after the initial e-mail, during this time one remainder was sent.

The two-section questionnaire was developed through several face-to-face and online meetings among the authors. The first section included questions on demographic and professional information of participants. In the second section, respondents were asked to rate, on a five-point scale ranging from 1 (“not important”) to 5 (“extremely important”), which items should be considered when assessing the overall quality of a NRS, specifically cohort and case-control studies, of pharmacological interventions.

In addition to the rating, the respondents had the option of providing any other relevant considerations at the end of the questionnaire (see Supplement 2).

The outcome variable of the online survey, originally consisting of five categories, was collapsed into two levels. Participants considered important an item if they rated it 4 or 5 (where 5 is "extremely important"), while they did not consider an item of key importance if they rated it 1, 2 or 3 (where 1 is "not important"). When at least 70% of participants agreed on the importance of an item (rating 1, 2 or 3 vs rating 4 or 5), we considered that a consensus had been reached on that item.

Thirty-five CER SIG experts participated to the first round of the survey (response rate = 73%). Eighteen respondents worked in academia (51%), 16 respondents worked in industry (46%), and only one was affiliated to a governmental institution (3%). The majority of respondents were from U.S. (n = 26, 74%), the others were from Europe (n = 7, 20%) or from Canada (n = 2, 6%). After participants ranked the relevance of the items and provided comments to the first questionnaire, we reviewed the responses and adjusted the survey to reflect the feedback. Overall, participants considered the presented domains important or extremely important. All domains, other than the domain n. 5 ("Lack of appropriateness of statistical analyses"), were rated relevant with unanimous consent (100% of participants rated them 4 or 5). For the domain n. 5 ("Lack of appropriateness of statistical analyses (with specific mention of adjustment for causal intermediaries, and/or incorrect outcome model specification)" in the Supplement 2), only three participants did not consider it important. Based on individual experts' suggestions, we added the subdomain "External validity of target population" to domain n. 1 "Methods of selecting participants" to address the external validity or generalizability of the study population to the eligible or target population. We removed "Recall bias" from the critical elements of domain n. 3 since this was the only domain to be rated not relevant by most of participants (n = 26, 74%). The rationale provided was that, since real world data (RWD) are often collected for intents unrelated to pursue research goals (e.g., administrative purposes), biases such as recall bias, interviewer bias, nonresponse bias are usually reduced or even eliminated [Sørensen HT et al.].

A revised survey was returned to the same participants for reassessment of the content. The response rate to the second round was 89% (n = 31/35). Since in this second round the threshold of 70% was reached for all the old and new items, we terminated the consensus procedure and communicated the results to the CER SIG members. The Delphi flowchart is shown in Figure S2, while the last version of cover letter and online questionnaire is available on the online Supplement 2.

The evaluation framework was built on the main domains and subdomains identified through the Delphi process and it was used as an instrument to evaluate the completeness of the existing assessment tools for cohort and case-control studies only or multiple NRS. The assessment of each domain and subdomain and the distribution of domains in the reviewed tools are shown in Figures S3-S11.

Table S1. Search strategy

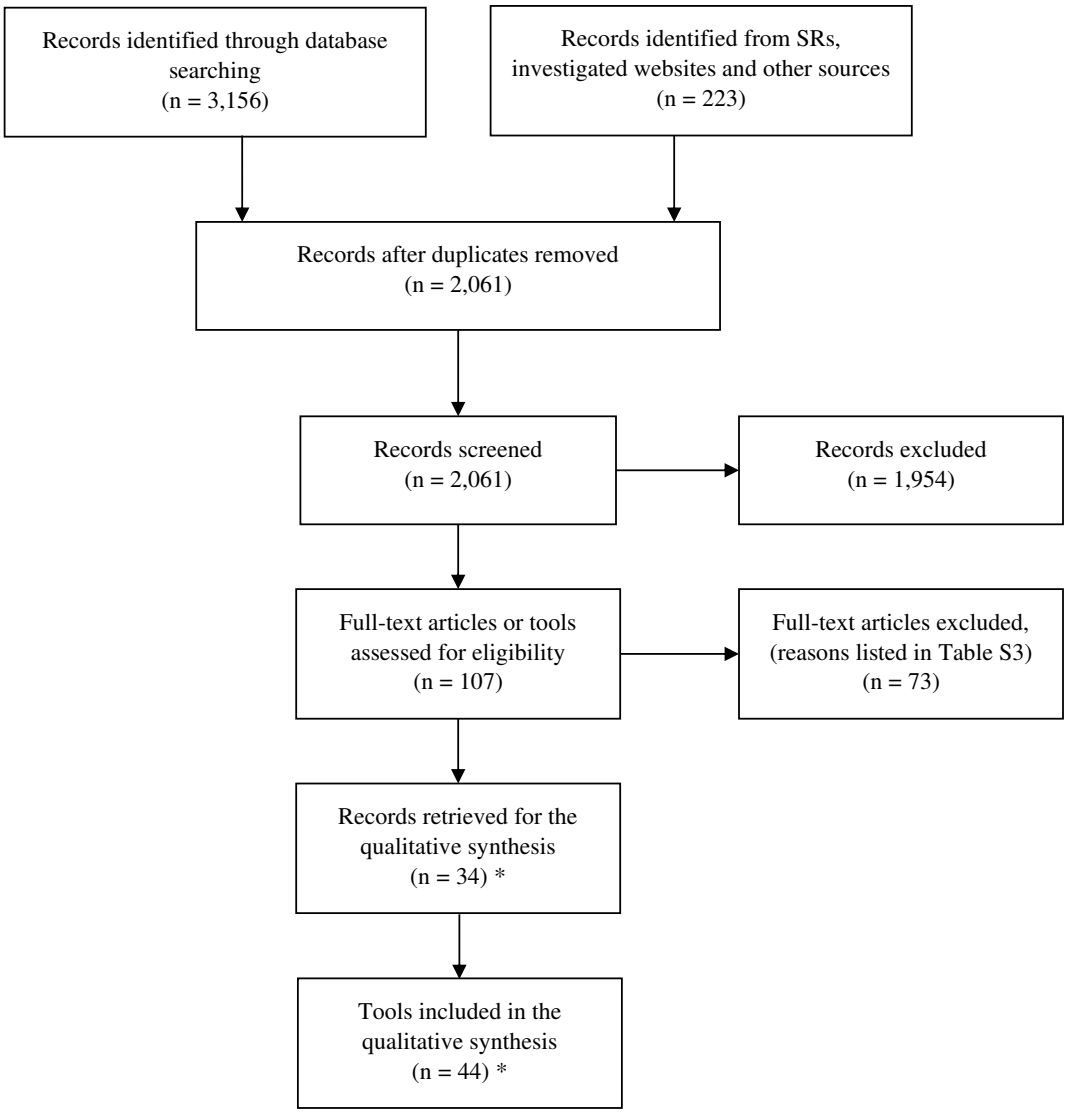
Items	N.	Terms
Critical appraisal tool	#1	"critical" [All Fields] AND "appraisal" [All Fields] AND "tools" [All Fields]
	#2	"critical" [All Fields] AND "appraisal" [All Fields]
	#3	("critical" [All Fields] AND "review" [All Fields]) OR "critical review" [All Fields]) AND form [All Fields]
	#4	("systematic review" [Publication Type] OR "systematic reviews as topic" [MeSH Terms] OR "systematic review"[All Fields]) AND form [All Fields]
	#5	appraisal [All Fields] AND ("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields] OR ("research" [All Fields] AND "methodology" [All Fields]) OR "research methodology"[All Fields])
	#6	("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields]) AND ("review" [Publication Type] OR "review literature as topic" [MeSH Terms] OR "review"[All Fields])
Study reporting tool	#7	"study" [All Fields] AND "reporting" [All Fields] AND "tool" [All Fields]
	#8	"study" [All Fields] AND "reporting" [All Fields]
	#9	"reporting" [All Fields] AND "form" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
	#10	"reporting" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
Tool	#11	"checklist" [MeSH Major Topic] OR "scale*" [Title/Abstract]
	#12	"surveys and questionnaires"[MeSH Major Topic] OR "questionnaire*" [Title/Abstract]
	#13	("tool*" [All Fields] OR "instrument*" [All Fields] OR "checklist*" [All Fields] OR "questionnaire*" [All Fields]) AND ("quality" [All Fields] OR "method*" [All Fields] OR "bias" [All Fields])
Study design	#14	"cohort studies"[MeSH Terms] OR cohort studies [Text Word] OR cohort stud* [All Fields]
	#15	"case-control studies" [MeSH Terms] OR case-control studies [Text Word] OR case control stud* [All Fields]
	#16	Non [All Fields] AND ("random allocation"[MeSH Terms] OR randomized [Text Word]) AND stud* [All Fields]
Systematic review	#17	"systematic review" [Publication Type] OR "systematic reviews as topic"[MeSH Terms] OR "systematic review"[All Fields]
Filters	#18	"humans"[MeSH Terms]
	#19	"Review" [ptyp] OR "systematic" [sb]
Strings		
1st search - tools*	#20	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (#14 OR #15 OR #16) AND #18
	#21	(#7 OR #8 OR #9 OR #10) AND (#14 OR #15 OR #16) AND #18
	#22	(#11 OR #12 OR #13) AND (#14 OR #15 OR #16) AND #18
2nd search - systematic reviews of tools*	#23	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#24	(#7 OR #8 OR #9 OR #10) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#25	(#11 OR #12 OR #13) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)

*tools are defined as instruments (e.g., qualitative checklists, questionnaires, scoring scales, etc.) that investigate the overall quality of a study, identifying potential biases – either used to critically appraise studies included in a systematic review, or to help in the peer-reviewing process of scientific publications – (i.e., critical appraisal tools), or that support the reporting of research methods and findings (i.e. study reporting tools).
Note: the strings were built in Medline and then adapted to Embase (through Elsevier)

Table S2. List of investigated websites of influential initiatives, programs or organizations.

Name of initiative, programs or institutions	**Link to the website**
STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) statement	https://strobe-statement.org/index.php?id=available-checklists
Effective Health Care (EHC) Program of the Agency for Healthcare Research and Quality (AHRQ)	https://effectivehealthcare.ahrq.gov/research-methods
The Ottawa Hospital – Research Institute	http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp
The Scottish Intercollegiate Guidelines Network (SIGN)	https://www.sign.ac.uk/checklists-and-notes.html
GRACE (Good ReseArch for Comparative Effectiveness) initiative	https://www.graceprinciples.org/about-grace.html
The Joanna Briggs Institute	http://joannabriggs.org/research/critical-appraisal-tools.html
TREND (Transparent Reporting of Evaluations with Nonrandomized Designs) statement	https://www.cdc.gov/trendstatement/index.html
CASP (Critical Appraisal Skills Programme)	https://casp-uk.net/casp-tools-checklists/

Figure S1. Flowchart of the selection of the tools



*Ten retrieved articles/records included two different assessment tools for case-control and cohort studies. Thus, the overall number of included tools raised up to 44.

The systematic reviews (SRs) were retrieved through a systematic search. Additional details about the SRs search are available upon request.

NB. The reporting items in the flowchart follows the indications given by the PRISMA Statement at Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097.

Figure S2. Flowchart of the Delphi procedure

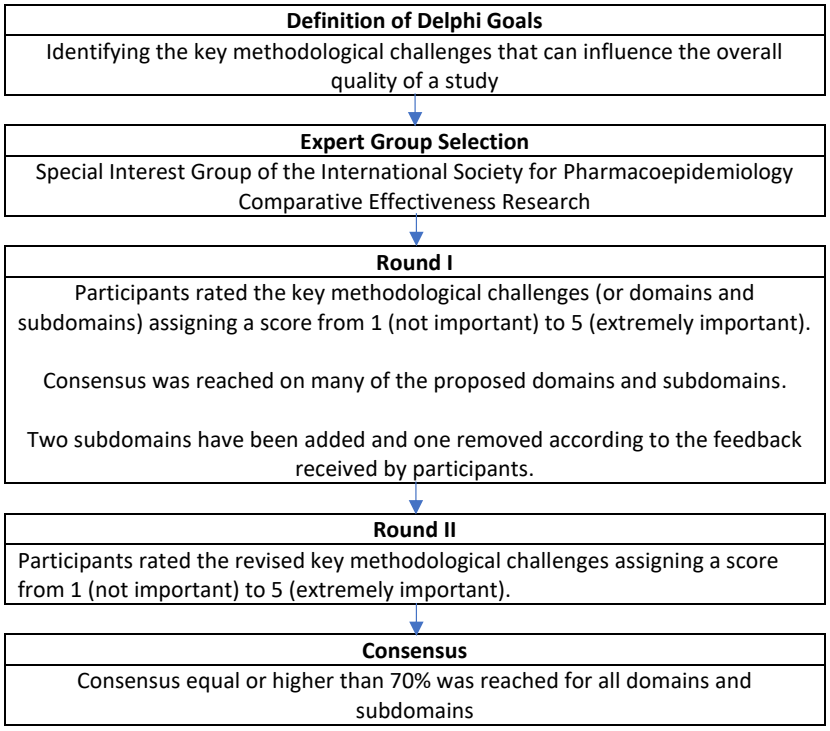


Table S3. Excluded tools after full-text screening

Reference (73 article excluded)	**Reason for exclusion**
Berkman ND, Lohr KN, Ansari M, et al. Chapter 15 Appendix A: A Tool for Evaluating the Risk of Reporting Bias (in Chapter 15: Grading the Strength of a Body of Evidence When Assessing Health Care Interventions for the Effective Health Care Program of the Agency for Healthcare Research and Quality: An Update). Methods Guide for Comparative Effectiveness Reviews (Prepared by the RTI-UNC Evidence-based Practice Center under Contract No. 290-2007-10056-I). AHRQ Publication No. 13(14)-EHC130-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of systematic reviews
Balshem H, Stevens A, Ansari M, et al. Finding grey literature evidence and assessing for outcome and analysis reporting biases when comparing medical interventions: AHRQ and the Effective Health Care Program. (Prepared by the Oregon Health and Science University and the University of Ottawa Evidence-based Practice Centers under Contract Nos. 290-2007-10057-I and 290-2007-10059-I.) AHRQ Publication No. 13(14)-EHC096-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of reporting bias, guidance
Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008; 336:924–6.	Assessment to provide strength of recommendations
Hayden JA van der Windt DA, Cartwright JL, et al. Assessing bias in studies of prognostic factors. Ann Intern Med 2013; 158:280–6.	Assessment of studies of prognostic factors
Higgins JPT, Altman DG, Sterne JAC. Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S, eds. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. London: The Cochrane Collaboration, 2011.	Assessment of RCTs
Higgins JPT, Savović J, Page MJ, et al. Revised Cochrane risk of bias tool for randomized trials (RoB 2.0), Version 20. 2016 Available at: https://sites. google.com/site/riskofbiastool/ (accessed 19 Sep 2019)	Assessment of RCTs
Hooijmans CR, Rovers MM, de Vries RB, et al. SYRCLE's risk of bias tool for animal studies. BMC Med Res Methodol 2014; 14:43	Assessment of animal studies
Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (MINORS): development and validation of a new instrument. ANZ J Surg. 2003;73(9): 712-716.	Assessment of NRS of surgical interventions
Centre for reviews and dissemination. Systematic Reviews: CRD's Guidance for Undertaking Reviews in Healthcare. 3rd ed. UK: University of York; 2009.	No assessment tool, guidance for reviews
Deeks JJ, Dinnes J, D'Amico R, et al. Evaluating non-randomised intervention studies. Health Technol. Assess (Winchester, England. 2003;7(27): iii-x):1-173.	No assessment tool, original research
NICE. Guide to the methods of technology appraisal 2013. Updated 2016.	No assessment tool, guidance for HTA
University of Wales College of Medicine. Critical Appraisal Forms. University of Wales, 2004.	N/A
Loney PL, Chambers LW, Bennett KJ, Roberts JG, Stratford PW. Critical appraisal of the health research literature: prevalence or incidence of a health problem. Chronic Dis Canada 2000;19:170–77.	No structured assessment tool
Cameron I, Crotty M, Currie C et al. Geriatric rehabilitation following fractures in older people: a systematic review. Health Technol Assess 2000;4: i–111.	No assessment tool
Carneiro AV. Critical appraisal of prognostic evidence: practical rules. Rev Port Cardiol 2002; 21:891–900.	Assessment tool of prognostic studies
Centre for Evidence-Based Mental Health, University of Oxford. Critical Appraisal Forms. 2004. Available at: http://www.cebmh.com/ (accessed 19 Sep 2019)	Assessment tool tailored to a particular disease/topic

Esdaile JM, Horwitz RI. Observational studies of cause-effect relationships: an analysis of methodologic problems as illustrated by the conflicting data for the role of oral contraceptives in the etiology of rheumatoid arthritis. J Chronic Dis 1986; 39:841–52.	No structured assessment tool
Khan KS, Riet GT, Popay J, Nixon J, Kleijnen J. Undertaking systematic reviews of research effectiveness. CRD's guidance for those carrying out or commissioning reviews. CRD Report number 4 (2nd edn). 2001. The University of York Centre for Reviews and Dissemination.	No structured assessment tool
How to read clinical journals: IV. To determine etiology or causation. Can Med Assoc J. 1981;124(8):985-990.	No structured assessment tool for NRS
Mulrow CD, Lichtenstein MJ. Blood glucose and diabetic retinopathy: a critical appraisal of new evidence. J Gen Intern Med 1986; 1:73–77.	Assessment tool tailored to a particular disease/topic
Whiting P, Rutjes AW, Reitsma JB, Bossuyt PM, Kleijnen J. The development of QUADAS: a tool for the quality assessment of studies of diagnostic accuracy included in systematic reviews. BMC Med Res Methodol 2003; 3:25.	Assessment tool for diagnostic accuracy studies
Campbell H, Rudan I. Interpretation of genetic association studies incomplex disease. Pharmacogenomics J 2002; 2:349–60.	Assessment tool tailored to a particular disease/topic
Solomon DH, Bates DW, Panush RS, Katz JN. Costs, outcomes, and patient satisfaction by provider type for patients with rheumatic and musculoskeletal conditions: a critical review of the literature and proposed methodologic standards. Ann Intern Med 1997; 127:52–6	Assessment tool tailored to a particular disease/topic
The STARD Group. The STARD Initiative—Towards Complete and Accurate Reporting of Studies on Diagnostic Accuracy. http://www.consort-statement.org/stardstatement.htm, 2001.	Assessment tool for diagnostic accuracy studies
Critical appraisal: Guidelines for the critical appraisal of a paper. Surgical-Tutor.org.uk, 2004.	N/A
Zola P, Volpe T, Castelli G et al. Is the published literature a reliable guide for deciding between alternative treatments for patients with early cervical cancer? Int J Radiat Oncol Biol Phys 1989;16:785–97.	No assessment tool, original research
Bollini P, Garcia Rodriguez LA, Gutthann SP, Walker AM. The impact of research quality and study design on epidemiologic estimates of the effect of nonsteroidal anti-inflammatory drugs on upper gastrointestinal tract disease. Arch Intern Med 1992; 152:1289–95.	No assessment tool, original research
Ciliska D, Hayward S, Thomas H et al. A systematic overview of the effectiveness of home visiting as a delivery strategy for public health nursing interventions. Can J Public Health 1996;87:193–98.	No assessment tool, original research
Cowley DE. Prostheses for primary total hip replacement. A critical appraisal of the literature. Int J Technol Assess Health Care 1995;11:770–78.	No assessment tool, original research
Spitzer WO, Lawrence V, Dales R et al. Links between passive smoking and disease: a best-evidence synthesis. A report of the Working Group on Passive Smoking. Clin Invest Med 1990;13: 17–42.	No assessment tool, original research
Steinberg EP, Eknoyan G, Levin NW et al. Methods used to evaluate the quality of evidence underlying the National Kidney Foundation- Dialysis Outcomes Quality Initiative Clinical Practice Guidelines: description, findings, and implications. Am J Kidney Dis 2000; 36:1–11.	No assessment tool, original research
Anders JF, Jacobson RM, Poland GA, Jacobsen SJ, Wollan PC. Secondary failure rates of measles vaccines: a meta-analysis of published studies. Pediatr Infect Dis J 1996; 15:62–66.	No assessment tool, original research
Ariens GA, van Mechelen W, Bongers PM, Bouter LM, van der WG. Physical risk factors for neck pain. Scand J Work Environ Health 2000; 26:7–19.	No assessment tool, original research
Berlin JA, Colditz GA. A meta-analysis of physical activity in the prevention of coronary heart disease. Am J Epidemiol 1990; 132:612–28.	No assessment tool, original research
Bhutta AT, Cleves MA, Casey PH, Cradock MM, Anand KJS. Cognitive and behavioral outcomes of school-aged children who were born preterm: a meta-analysis. J Am Med Assoc 2002;288: 728–37.	No assessment tool, original research

Borghouts JA, Koes BW, Bouter LM. The clinical course and prognostic factors of non-specific neck pain: a systematic review. Pain 1998; 77:1–13.	No assessment tool, original research
Campos-Outcalt D, Senf J, Watkins AJ, Bastacky S. The effects of medical school curricula, faculty role models, and biomedical research support on choice of generalist physician careers: a review and quality assessment of the literature. Acad Med 1995; 70:611–19.	No assessment tool, original research
Carson CA, Fine MJ, Smith MA, Weissfeld LA, Huber JT, Kapoor WN. Quality of published reports of the prognosis of community-acquired pneumonia. J Gen Intern Med 1994; 9:13–19.	No assessment tool, original research
Corrao G, Bagnardi V, Zambon A, Arico S. Exploring the doseresponse relationship between alcohol consumption and the risk of several alcohol-related conditions: a meta-analysis. Addiction 1999; 94:1551–73.	No assessment tool, original research
Garber BG, Hebert PC, Yelle JD, Hodder RV, McGowan J. Adult respiratory distress syndrome: a systemic overview of incidence and risk factors. Crit Care Med 1996; 24:687–95.	No assessment tool, original research
Goodman SN, Berlin J, Fletcher SW, Fletcher RH. Manuscript quality before and after peer review and editing at Annals of Internal Medicine. Ann Intern Med 1994; 121:11–21.	No assessment tool
Jabbour M, Osmond MH, Klassen TP. Life support courses: are they effective? Ann Emerg Med 1996; 28:690–98.	No assessment tool, original research
Kreulen CM, Creugers NH, Meijering AC. Meta-analysis of anterior veneer restorations in clinical studies. J Dent 1998; 26:345–53.	No assessment tool, original research
Krogh CL. A checklist system for critical review of medical literature. Med Educ 1985; 19:392–95.	No assessment tool for NRS validity
Littenberg B, Weinstein LP, McCarren M et al. Closed fractures of the tibial shaft. A meta-analysis of three methods of treatment. J Bone Joint Surg Am 1998; 80:174–83.	No assessment tool, original research
Longnecker MP, Berlin JA, Orza MJ, Chalmers TC. A meta-analysis of alcohol consumption in relation to risk of breast cancer. JAMA 1988; 260:652–56.	No assessment tool, original research
Macfarlane TV, Glenny AM, Worthington HV. Systematic review of population-based epidemiological studies of oro-facial pain. J Dent 2001;29:451–67.	No assessment tool, original research
Manchikanti L, Singh V, Vilims BD, Hansen HC, Schultz DM, Kloth DS. Medial branch neurotomy in management of chronic spinal pain: systematic review of the evidence. Pain Physician 2002;5:405–18.	No assessment tool, original research
Meijer R, Ihnenfeldt DS, van Limbeek J, Vermeulen M, de Haan RJ. Prognostic factors in the subacute phase after stroke for the future residence after six months to one year. A systematic review of the literature. Clin Rehabil 2003;17:512–20.	No assessment tool, original research
Nguyen QV, Bezemer PD, Habets L, Prahl-Andersen B. A systematic review of the relationship between overjet size and traumatic dental injuries. Eur J Orthod 1999;21:503–15.	No assessment tool, original research
Rangel SJ, Kelsey J, Colby CE, Anderson J, Moss RL. Development of a quality assessment scale for retrospective clinical studies in pediatric surgery. J Pediatr Surg 2003;38:390–96.	Assessment tool tailored to a particular disease/topic
Reisch JS, Tyson JE, Mize SG. Aid to the evaluation of therapeutic studies. Pediatrics 1989;84:815–27.	No assessment tool
Stock SR. Workplace ergonomic factors and the development of musculoskeletal disorders of the neck and upper limbs: a metaanalysis. Am J Ind Med 1991;19:87–107.	No assessment tool, original research
van der Windt DAWM, Thomas E, Pope DP et al. Occupational risk factors for shoulder pain: a systematic review. Occup Environ Med 2000;57:433–42.	No assessment tool, original research
Margetts BM, Vorster HH, Venter CS. Evidence-based nutrition—review of nutritional epidemiological studies. South African J Clin Nutr 2002;15:68–73.	Assessment tool tailored to a particular disease/topic
Higgins JP, Altman DG, Gotzsche PC, Juni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011; 343: d5928.	Assessment of RCTs
Campos TF, Beckenkamp PR, Moseley AM. Usage evaluation of a resource to support evidence-based physiotherapy: the physiotherapy evidence database (PEDro). Physiotherapy 2013; 99(3): 252–57.	Assessment of RCTs

Jadad AR, Moore RA, Carroll D, Jenkinson C, Reynolds DJ, Gavaghan DJ, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Controlled Clinical Trials 1996; 17(1): 1–12	Assessment of RCTs
Verhagen AP, de Vet HC, de Bie RA, Kessels AG, BoersM, Bouter LM, et al. The Delphi list: a criteria list for quality assessment of randomized clinical trials for conducting systematic reviews developed by Delphi consensus. Journal of Clinical Epidemiology 1998; 51(12): 1235–41.	Assessment of RCTs
Moga C, Guo B, Schopflocher D, Harstall C. Development of a Quality Appraisal Tool for Case Series Studies using a Modified Delphi Technique. 2012.	Assessment of case series studies
Stroke Therapy Academic Industry R. Recommendations for standards regarding preclinical neuroprotective and restorative drug development. Stroke 1999; 30(12): 2752–58.	Assessment of animal studies
Fisher M, Feuerstein G, Howells DW, Hurn PD, Kent TA, Savitz SI, et al. Update of the stroke therapy academic industry roundtable preclinical recommendations. Stroke 2009; 40(6): 2244–50.	Assessment of animal studies
Macleod MR, O'Collins T, Howells DW, Donnan GA. Pooling of animal experimental data reveals influence of study design and publication bias. Stroke 2004; 35(5): 1203–08.	Assessment of animal studies
Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Medical Research Methodology 2007; 7: 10.	Assessment of systematic reviews
Collaboration A. Development and validation of an international appraisal instrument for assessing the quality of clinical practice guidelines: the AGREE project. Quality & Safety in Health Care 2003; 12(1): 18–23	Assessment of clinical practice guidelines
Brouwers MC, Kho ME, Browman GP, Burgers JS, Cluzeau F, Feder G, et al. AGREE II: advancing guideline development, reporting and evaluation in health care. Canadian Medical Association Journal 2010; 182(18): E839–E842.	Assessment of clinical practice guidelines
Kirkham JJ, Dwan KM, Altman DG, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ 2010;340:c365.	Assessment of RCTs
Meader N, King K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. Systematic reviews 2014;3(1):82.	Assessment of systematic reviews
Dwan K, Gamble C, Kolamunnage-Dona R, et al. Assessing the potential for outcome reporting bias in a review: a tutorial. Trials 2010;11:52.	Assessment of RCTs
Stewart GB, Higgins JP, Schunemann H, et al. The use of Bayesian networks to assess the quality of evidence from research synthesis: 1. PLoS One 2015;10(3):e0114497.	Assessment of systematic reviews
Salanti G, Giovane CD, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. PLoS One 2014;9(7):e99682.	Assessment for network meta-analyses
Higgins JP, Del Giovane C, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. Value Health 2014;17(7):A324.	Assessment for network meta-analyses
WP5, EMA, EUnetHTA Joint Action 3 (2016 - 2020), REQueST® tool. Available at: https://eunethta.eu/wp-content/uploads/2019/10/EUnetHTAJA3_Vision_paper-on-REQueST-tool.pdf	Assessment specific on the quality of registries

Figure S3. Assessment tools that included or not at least one question/item to address subdomains of methods for selecting study participants (Domain 1), stratified by study design.

	Domain 1 - Methods for selecting participants (SELECTION BIAS)				
Tools for Case-Control Studies only	Sampling strategies to correct selection bias	Inclusion and exclusion criteria of target population	Depletion of susceptibles	Generalizability of target population	Others^
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	✓	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	×	×	×	✓
NIH–NHLBI 2014	✓	✓	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	✓	×	✓	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Durant 1994	×	×	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Lichtenstein 1987	✓	✓	×	×	✓
Horwitz 1979	✓	✓	×	×	✓
Tools for Cohort Studies only					
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	×	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	✓	×	×	✓
NIH–NHLBI 2014	×	✓	×	✓	✓
HEB Wales 2014	×	×	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	×	×	×	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Hadorn 1996	✓	✓	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Tools for multiple NRS					
RELEVANT 2019	×	✓	×	✓	✓
MMAT 2018	×	✓	×	✓	×
ROBINS-I 2016	×	✓	✓	×	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	×	✓
GRACE 2014 (#)	×	×	×	×	×
RoBANS 2013	×	×	×	×	✓
RTI 2013	✓	✓	×	×	✓
Montreal 2011	✓	×	×	✓	×
EPHPP 2004	×	×	×	✓	✓
TREND statement 2004	✓	✓	×	×	✓
Margetts 2002	✓	✓	×	×	✓
Zaza 2000	✓	✓	×	✓	✓
Downs-Black 1998	×	✓	×	✓	✓
Elwood 1998	×	✓	×	×	✓
London 1996	×	×	×	✓	✓
Avis 1994	✓	×	×	×	✓
Levine 1994	×	×	×	×	✓
Cho 1994 (#)	×	✓	×	×	✓
COEH 1991	✓	✓	×	✓	✓
Fowkes-Fulton 1991	✓	✓	×	✓	✓
Gardner 1986	×	×	×	×	✓

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
^ "Others" refers to items not included in our evaluation framework but included in the reviewed tools to investigate selection bias
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

Figure S4. Assessment tools that included or not at least one question/item to address subdomains for measurement of exposure, outcomes, covariate variables and follow-up (Domain 2), stratified by study design.

	Domain 2 - Measurement of exposure, outcomes, covariate variables and follow-up.			
Tools for Case-Control Studies only	Exposure	Outcome	Covariate variables	Follow-up
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	×
SURE 2018	✓	✓	×	×
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	×
Newcastle-Ottawa 2013	✓	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Durant 1994	×	×	✓	×
Gyorkos 1994	✓	✓	×	×
Lichtenstein 1987	✓	✓	×	×
Horwitz 1979	✓	✓	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	✓
SURE 2018	✓	✓	×	✓
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	✓
HEB Wales 2014	✓	✓	✓	✓
Newcastle-Ottawa 2013	✓	✓	✓	✓
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Hadorn 1996	✓	✓	×	×
Gyorkos 1994	✓	✓	×	✓
Tools for multiple NRS				
RELEVANT 2019	✓	✓	×	×
MMAT 2018	✓	✓	×	×
ROBINS-I 2016	✓	✓	✓	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	✓
GRACE 2014 (#)	✓	✓	✓	×
RoBANS 2013	✓	✓	×	×
RTI 2013	✓	✓	✓	✓
Montreal 2011	×	✓	✓	×
EPHPP 2004	✓	✓	×	×
TREND statement 2004	✓	✓	×	✓
Margetts 2002	✓	✓	×	×
Zaza 2000	✓	✓	×	×
Downs-Black 1998	✓	✓	×	×
Elwood 1998	✓	✓	×	×
London 1996	✓	✓	×	×
Avis 1994	✓	✓	×	×
Levine 1994	✓	✓	×	✓
Cho 1994 (#)	×	×	×	×
COEH 1991	✓	×	×	×
Fowkes-Fulton 1991	✓	✓	×	×
Gardner 1986	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

Figure S5. Assessment tools that included or not at least one question/item to address subdomains to detect design-specific sources of biases (Domain 3), stratified by study design.

	Domain 3 - Design-specific sources of bias						
Tools for Case-Control Studies only	new user design	active comparator design	immortal time bias	detection or surveillance bias	loss to follow-up bias	Non-contemporaneous comparator bias	Reverse causation
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	×	×	×
CASP 2018	×	×	×	×	×	×	×
SURE 2018	×	×	×	×	×	×	×
JBI 2017	×	×	×	×	×	×	×
NIH–NHLBI 2014	×	×	×	×	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	×	✓	×	×
SIGN Version 3.0 2012	×	×	×	×	×	×	×
STROBE statement V4 2007	×	×	×	×	×	×	×
Durant 1994	×	×	×	✓	×	×	×
Gyorkos 1994	×	×	×	×	×	×	×
Lichtenstein 1987	×	×	×	×	×	×	×
Horwitz 1979	×	×	×	✓	×	×	×
Tools for Cohort Studies only							
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	✓	×	✓
CASP 2018	×	×	×	×	×	×	×
SURE 2018	×	×	×	×	✓	×	×
JBI 2017	×	×	×	×	✓	×	✓
NIH–NHLBI 2014	×	×	×	×	✓	×	✓
HEB Wales 2014	×	×	×	×	✓	×	×
Newcastle-Ottawa 2013	×	×	×	×	✓	×	✓
SIGN Version 3.0 2012	×	×	×	✓	✓	×	✓
STROBE statement V4 2007	×	×	×	×	✓	×	×
Hadorn 1996	×	×	×	×	✓	×	×
Gyorkos 1994	×	×	×	×	×	×	×
Tools for multiple NRS							
RELEVANT 2019	×	×	×	×	×	×	×
MMAT 2018	×	×	×	×	✓	×	×
ROBINS-I 2016	✓	×	✓	×	✓	✓	✓
ISPOR-AMCP-NPC 2014 (#)	×	×	×	×	×	✓	×
GRACE 2014 (#)	✓	×	✓	×	×	✓	×
RoBANS 2013	×	×	×	✓	✓	✓	×
RTI 2013	×	×	×	×	✓	×	×
Montreal 2011	×	×	×	×	✓	×	×
EPHPP 2004	×	×	×	×	✓	×	×
TREND statement 2004	×	×	×	×	×	×	×
Margetts 2002	×	×	×	×	×	×	×
Zaza 2000	×	×	×	×	✓	×	×
Downs-Black 1998	×	×	×	×	✓	✓	×
Elwood 1998	×	×	×	×	×	×	✓
London 1996	×	×	✓	×	✓	×	×
Avis 1994	×	×	×	×	✓	×	×
Levine 1994	×	×	×	×	×	×	×
Cho 1994 (#)	×	×	×	×	✓	×	×
COEH 1991	×	×	×	×	×	×	×
Fowkes-Fulton 1991	×	×	×	×	✓	×	×
Gardner 1986	×	×	×	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

Figure S6. Assessment tools that included or not at least one question/item to address confounding (Domain 4), stratified by study design.

	Domain 4 - Confounding		
Tools for Case-Control Studies only	study design used to minimize the confounding	confounders measured and included in the analyses	potential unmeasured confounding addressed in the analyses
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Durant 1994	✓	✓	✓
Gyorkos 1994	×	✓	×
Lichtenstein 1987	✓	✓	×
Horwitz 1979	×	×	×
Tools for Cohort Studies only			
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
HEB Wales 2014	×	✓	✓
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Hadorn 1996	✓	✓	×
Gyorkos 1994	×	✓	×
Tools for multiple NRS			
RELEVANT 2019	✓	✓	×
MMAT 2018	✓	✓	×
ROBINS-I 2016	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	✓
GRACE 2014 (#)	✓	✓	×
RoBANS 2013	✓	✓	×
RTI 2013	✓	✓	×
Montreal 2011	×	✓	×
EPHPP 2004	✓	✓	×
TREND statement 2004	✓	✓	×
Margetts 2002	×	×	×
Zaza 2000	✓	✓	✓
Downs-Black 1998	×	✓	×
Elwood 1998	✓	✓	×
London 1996	✓	✓	×
Avis 1994	×	✓	×
Levine 1994	×	×	×
Cho 1994 (#)	✓	✓	×
COEH 1991	×	✓	×
Fowkes-Fulton 1991	✓	✓	×
Gardner 1986	×	×	×

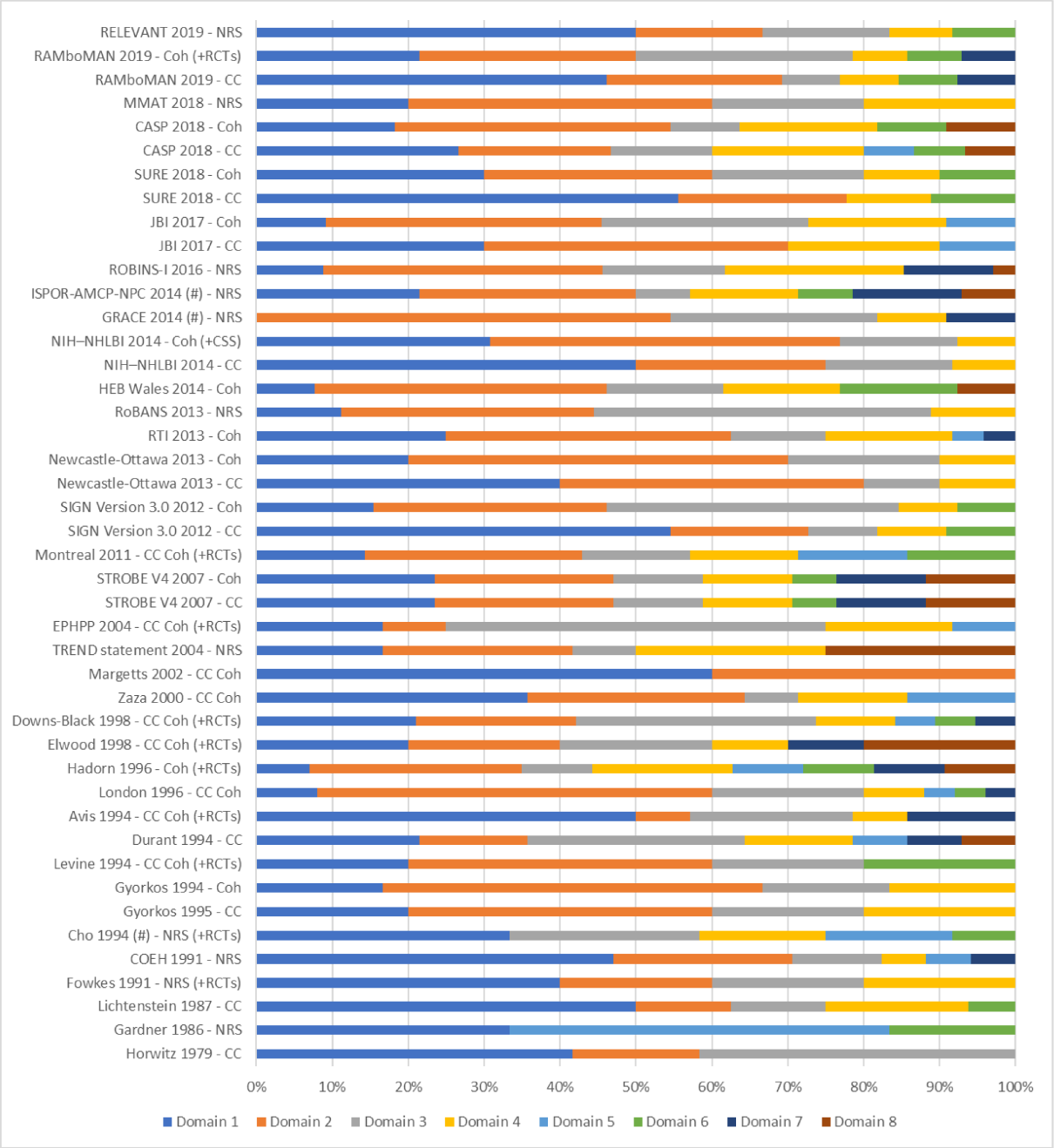
Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

	Domain 5 - Lack of appropriate statistical analyses	**Domain 6 - Statistical uncertainty**	**Domain 7 - Internal validity**	**Domain 8 - External validity**
Tools for Case-Control Studies only	(e.g., overadjustment for intermidiates, incorrect outcome model specification)	(e.g., confidence intervals reported for each analysis)	(e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)	(e.g. post-hoc subgroup analysis, comparison with other populations etc.)
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	✓	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Durant 1994	✓	×	✓	✓
Gyorkos 1994	×	×	×	×
Lichtenstein 1987	×	✓	×	×
Horwitz 1979	×	×	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	×	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
HEB Wales 2014	×	✓	×	✓
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Hadorn 1996	✓	✓	✓	✓
Gyorkos 1994	×	×	×	×
Tools for multiple NRS				
RELEVANT 2019	×	✓	×	×
MMAT 2018	×	×	×	×
ROBINS-I 2016	✓	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	×	✓	✓	✓
GRACE 2014 (#)	×	×	✓	×
RoBANS 2013	×	×	×	×
RTI 2013	✓	×	✓	×
Montreal 2011	✓	✓	×	×
EPHPP 2004	✓	×	×	×
TREND statement 2004	×	×	×	✓
Margetts 2002	×	×	×	×
Zaza 2000	✓	×	×	×
Downs-Black 1998	✓	✓	✓	×
Elwood 1998	×	×	✓	✓
London 1996	✓	✓	✓	×
Avis 1994	×	×	✓	×
Levine 1994	×	✓	×	×
Cho 1994 (#)	✓	✓	×	×
COEH 1991	✓	×	✓	×
Fowkes-Fulton 1991	×	×	×	×
Gardner 1986	✓	✓	×	×

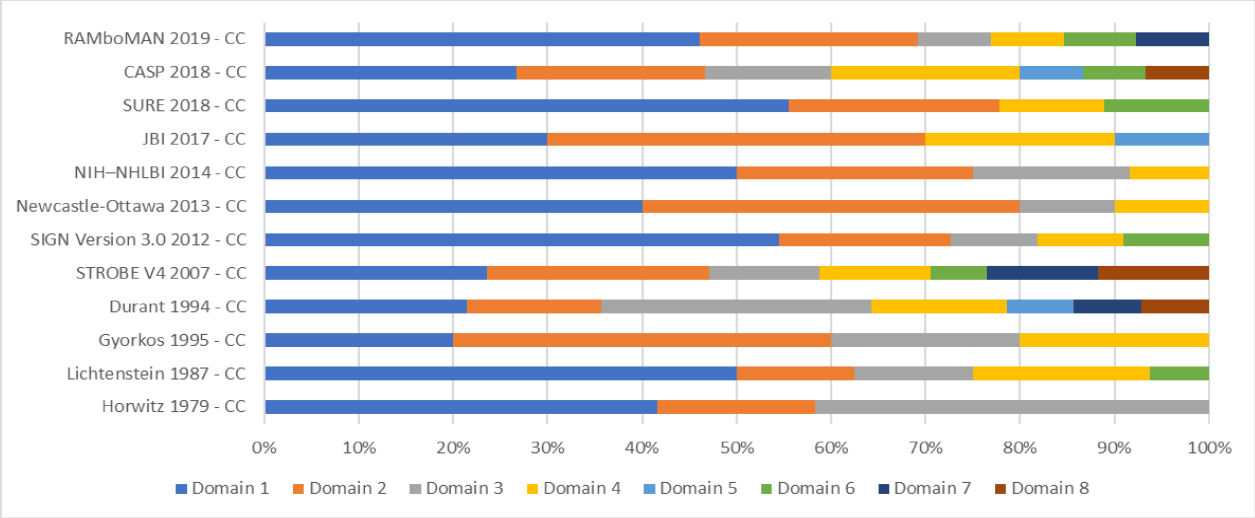
Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

Figure S8. Proportion of domains covered by each reviewed tool



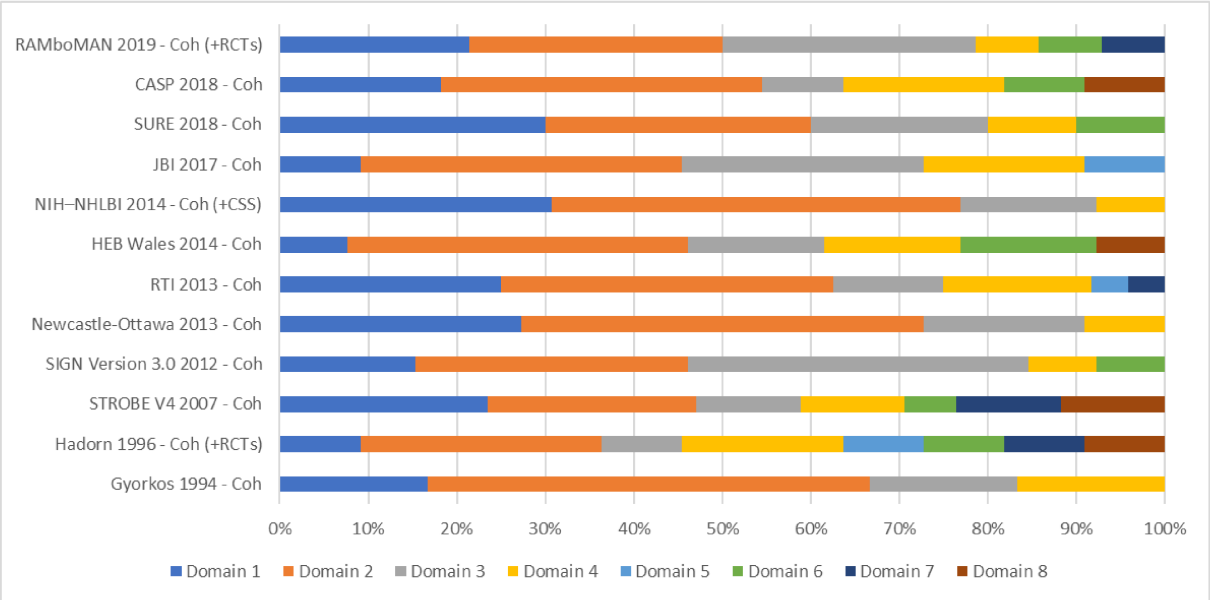
Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

Figure S9. Proportion of domains covered by each reviewed tool of case-control studies



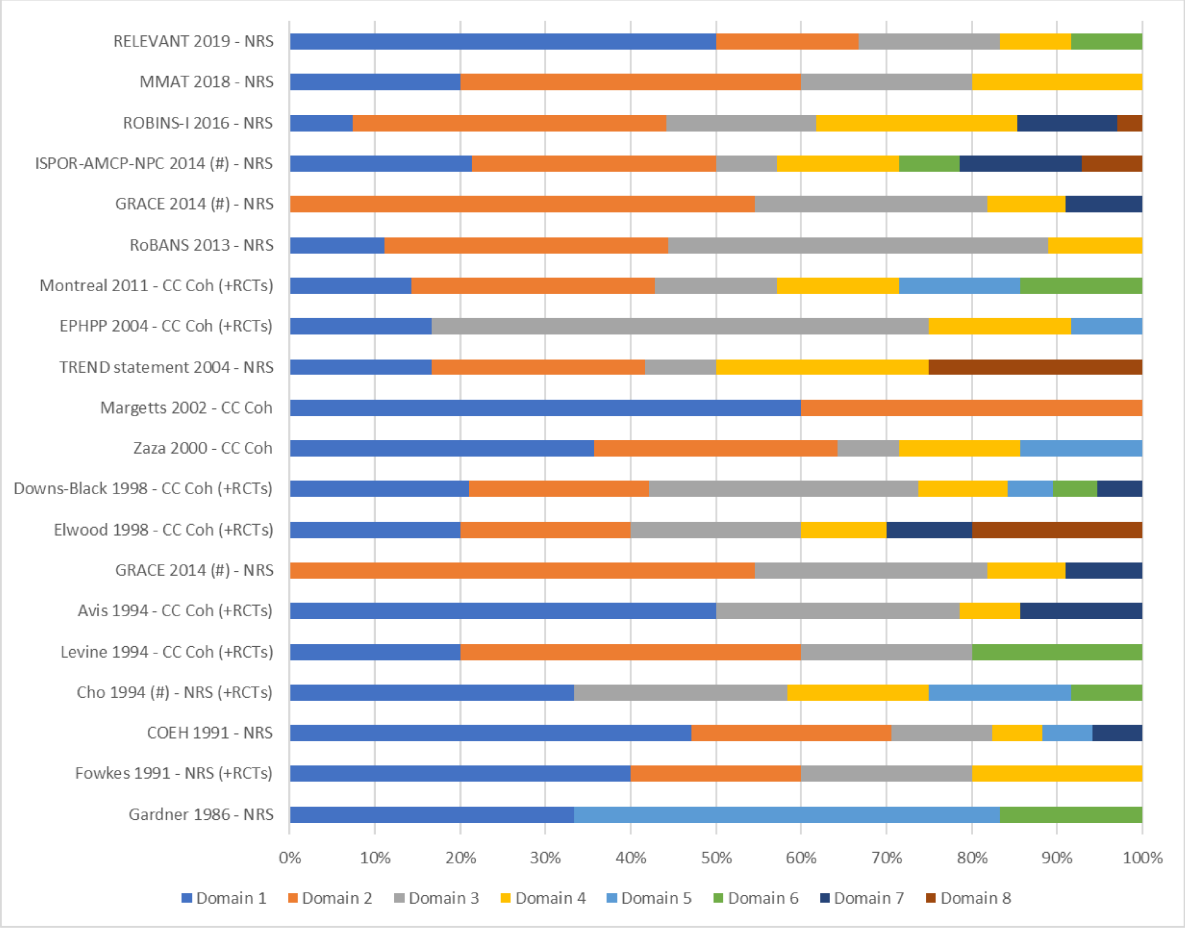
Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

Figure S10. Proportion of domains covered by each reviewed tool of cohort studies



Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

Figure S11. Proportion of domains covered by each reviewed tool of multiple study designs (NRS)



Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

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